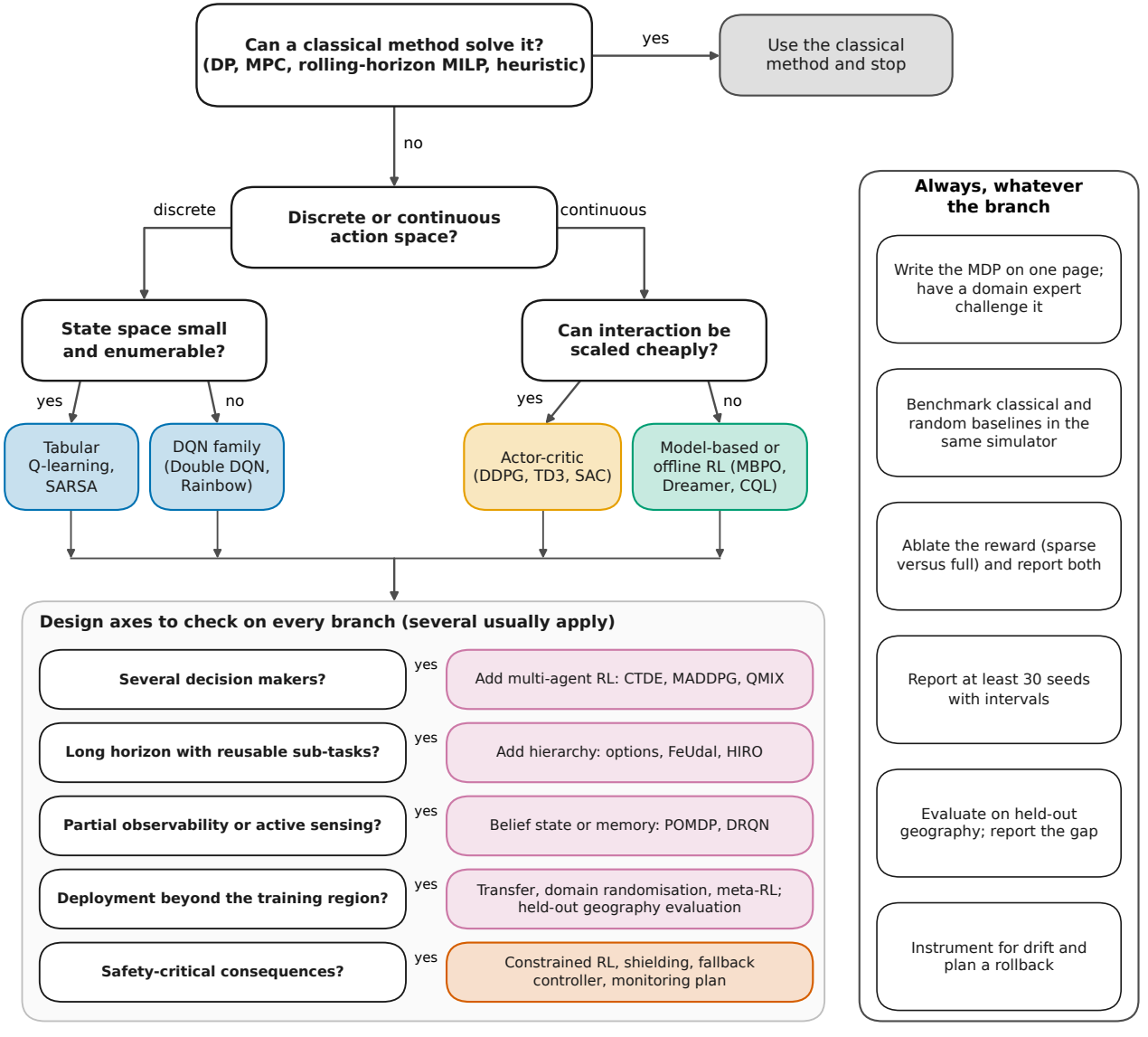
Can a classical method solve it? (DP, MPC, rolling-horizon MILP, heuristic)
yes
Use the classical method and stop
no
discrete
Discrete or continuous action space?
continuous
State space small and enumerable?
Can interaction be scaled cheaply?
yes
no
yes
no
Tabular Q-learning, SARSA
DQN family (Double DQN, Rainbow)
Actor-critic (DDPG, TD3, SAC)
Model-based or offline RL (MBPO, Dreamer, CQL)
Design axes to check on every branch (several usually apply)
Several decision makers?
yes
Add multi-agent RL: CTDE, MADDPG, QMIX
Long horizon with reusable sub-tasks?
yes
Add hierarchy: options, FeUdal, HIRO
Partial observability or active sensing?
yes
Belief state or memory: POMDP, DRQN
Deployment beyond the training region?
yes
Transfer, domain randomisation, meta-RL; held-out geography evaluation
Safety-critical consequences?
yes
Constrained RL, shielding, fallback controller, monitoring plan
Always, whatever the branch
Write the MDP on one page; have a domain expert challenge it
Benchmark classical and random baselines in the same simulator
Ablate the reward (sparse versus full) and report both
Report at least 30 seeds with intervals
Evaluate on held-out geography; report the gap
Instrument for drift and plan a rollback